

Why this note exists

Built while constructing `stage2_data` for the stage-2 ELVIS estimator: the single- τ model's "benefit shifter" $\tau\rho_t$ was being proxied by the *corrected effective sales tax rate* $\tau_{\tilde{C},it} = \tau_{S,it} - \tau_{P,it}\beta$ (600-opt-tax.qmd's "What about τ ?" slide). That proxy is ≤ 0 for **26.5%** of unincorp firm-periods (stable ~ 24 – 29% every year 1981–91, not an early "pre-tax era" artifact) — meaning $h(e) = \ln(\tau\rho_t) + \dots$ is undefined for over a quarter of the estimation sample under that proxy.

If I use a single tax rate in the model that captures the effective sales tax rate (the ratio of the difference in sales taxes paid on sales and the sales taxes paid on purchases over sales), then the marginal benefit in the model becomes undefined for more than a quarter of the unincorporated firms that have a negative rate. We lose a quarter of observations going in this direction.

A negative $\tau_{\tilde{C},it}$ means the (β -scaled) tax credit claimed on purchases exceeds the tax owed on sales. In this case, the firm is due a **refund** from the tax authority. Firms in industries exempted from paying sales taxes on their sales, such as food industry, commonly had negative tax rates because they paid taxes on their materials inputs.

Since the data separately report τ_S (sales tax rate on sales) and τ_P (sales tax rate on purchases), we can incorporate these data by allowing our model to have $\tau_S \neq \tau_P$. I derived this model below.

The two-tax profit function

Firm buys the true quantity M of materials at price ρ_t (cash cost $\rho_t M$), but *reports* $M^* = M + e$ to the tax authority in order to claim a purchases-tax credit at rate τ_P on the reported amount. It sells Y at price P_t and owes sales tax at rate τ_S on that revenue. If audited (probability $q(e)$, "always caught" upon audit, extra penalty multiplier $1 + \phi$ on the fraudulently claimed credit), the excess credit on the e overreported units must be repaid. Expected profit, net of the evasion cost $\kappa(e, \omega)$:

$$\begin{aligned} \pi = & (1 - q(e)) [P_t E_\varepsilon[Y_{it}] - \rho_t M_{it} - (\tau_S P_t Y_{it} - \tau_P \rho_t (M_{it} + e_{it}))] \\ & + q(e) [P_t E_\varepsilon[Y_{it}] - \rho_t M_{it} - (\tau_S P_t Y_{it} - \tau_P \rho_t M_{it}) - \phi \rho_t e_{it}] \\ & - \kappa_{it}(e_{it}, \omega_{it}) \end{aligned}$$

Simplifying,

$$\pi = \underbrace{(1 - \tau_S)P_t Y}_{\text{after-tax revenue}} - \underbrace{\rho_t M}_{\text{cash cost of true purchases}} + \underbrace{\tau_P \rho_t (M + e)}_{\text{credit claimed on reported purchases}} - \underbrace{q(e)(1 + \phi) \tau_P \rho_t e}_{\text{expected clawback}} - \kappa(e, \omega)$$

Collecting terms:

$$\pi = (1 - \tau_S)P_t Y - (1 - \tau_P)\rho_t M + \tau_P \rho_t e [1 - (1 + \phi)q(e)] - \kappa(e, \omega) \quad (1)$$

This model nests the single- τ version. By setting $\tau_S = \tau_P = \tau$ collapses Equation 1 to $P_t Y(1 - \tau) - \rho_t M(1 - \tau) + \tau \rho_t e [1 - (1 + \phi)q(e)] - \kappa$, the same structure as the previous model.

Evasion FOC: only τ_P survives

$\partial\pi/\partial e$: the $(1 - \tau_S)P_t Y$ term doesn't depend on e at all and drops out entirely.

$$\tau_P \rho_t [1 - (1 + \phi)(q + q'e)] = \frac{\partial\kappa}{\partial e} \quad (2)$$

This FOC indicates that the relevant rate is the sales tax on purchases. Identical in form to the model's existing evasion FOC, with the benefit-shifter now unambiguously $\tau_P \rho_t$. Under the old single- τ assumption this was indistinguishable from $\tau_S \rho_t$ or any combination, but with two rates it is **only** τ_P . $\tau_{S,it}$ (`sales_tax_rate_sales`) never belongs in $h(e)$ at all; $\tau_{P,it}$ (`sales_tax_rate_purchases`) does, directly, with no correction or combination needed.

Comparative static, solved explicitly. Under the headline linear- q spec ($q = \lambda e$, $\phi = 0$, κ linear in e so $\partial\kappa/\partial e = C(\omega, \psi)$ is constant in e), Equation 2 solves in closed form:

$$e^* = \frac{1}{2\lambda} - \frac{C(\omega, \psi)}{2\lambda \tau_P \rho_t} \quad \Rightarrow \quad \frac{\partial e^*}{\partial \tau_P} = \frac{C(\omega, \psi)}{2\lambda \rho_t \tau_P^2} > 0.$$

Higher τ_P increases optimal evasion. This is not an artifact of the linear- q functional form: by the implicit function theorem, $\partial e^*/\partial \tau_P = -F_{\tau_P}/F_e$ where $F_{\tau_P} = \kappa'(e^*)/\tau_P > 0$ (directly from the FOC, since $\kappa' > 0$) and $F_e < 0$ is exactly the second-order condition for e^* to be a maximum — so the sign is forced by the SOC holding at all, for any $q(\cdot)$. Intuition for why this isn't "more gets clawed back if caught, so the net benefit falls" (an earlier, wrong reading in this session): the clawback term $(1 + \phi)(q + q'e)$ scales with e , not with τ_P — raising τ_P scales up the *entire* bracket, kept portion and clawback-risk portion together, uniformly. Every unit of e becomes more valuable whether or not the firm is caught, so the firm optimally overreports more, up to where the rising detection probability catches up.

A regression of $\mathcal{V}_{it}$ on $\tau_{P,it}$ cannot test this — mechanical confound, not a sign check. `sales_tax_rate_purchases = t2/s10` (Code/Colombia/10_data_wrangling.R:131), where `s10=nom_mats=` nominal M_{it}^* ; `log_mats_share = log(nom_mats/nom_gross_output)` (line 275) also uses M_{it}^* , in the numerator. So $\mathcal{V}_{it}$ and $\tau_{P,it}$ share M_{it}^* in opposite positions: any firm that overreports more shows a mechanically higher $\mathcal{V}_{it}$ *and* a mechanically lower $\tau_{P,it}$ (same tax paid, divided by an inflated M^*), independent of the true economics. This is the same class of problem `CLAUDE.md` already flags for regressing $\mathcal{V}_{it}$ on M_{it}^* directly. Confirmed empirically: `feols(cal_V`

`~ sales_tax_rate_purchases | sic_3+year)` gives -1.890^{***} ; controlling for `sales_tax_rate_sales` jointly still gives -2.025^{***} on the purchases rate — the mechanical contamination survives conditioning on τ_S . **None of these regressions are valid evidence against the derivation above** — a clean test would need the same fix already prescribed for the $\mathcal{V} \sim M^*$ problem (instrument or bucket by something that doesn't enter $\mathcal{V}_{it}$'s own construction, e.g. K_{it} or L_{it}), not attempted here.

Materials FOC: the tax wedge

$\partial\pi/\partial M$ (through $Y = K^{\alpha_K} L^{\alpha_L} M^\beta e^{\omega+\varepsilon}$), holding e fixed so $\partial M^*/\partial M = 1$ contributes nothing new — the $\tau_P \rho_t e$ term has zero derivative in M once e is held fixed:

$$(1 - \tau_S) P_t \beta \frac{Y}{M} = \rho_t (1 - \tau_P) \implies \frac{\rho_t M_{it}}{P_t Y_{it}} = \beta \cdot \frac{1 - \tau_{S,it}}{1 - \tau_{P,it}} \cdot e^{-\varepsilon_{it}} \quad (3)$$

$M \perp e$ is preserved exactly (e drops out of this FOC entirely, same as before) — but the materials FOC now carries a multiplicative tax wedge $(1 - \tau_S)/(1 - \tau_P)$ that the model's documented $\varepsilon_{it}(M_{it}) = \ln \beta - \ln(\rho_t M_{it}/P_t Y_{it})$ (9999-elvis.qmd, eq-full-eps) does not have. This is the consequential part: $\varepsilon(M)$ is the single most load-bearing map in the whole ELVIS construction, feeding $\omega(M)$, $h(e)$, every moment — and it also underlies **stage 1's own regression** (`first_stage_panel_me` regresses `log_mats_share` on a constant among *corp* firms only, whose residual is $-\varepsilon$). Corp firms face the same varying τ_S, τ_P as everyone else — evasion status is irrelevant to which tax rates apply — so if the wedge varies across corp observations, it is currently an **omitted variable** contaminating $\hat{\beta}$ and the recovered $\hat{\varepsilon}$ residuals, not a stage-2-only issue.

The tax authority have real effects on firms' optimality input decision, by changing either the sales tax on sales or purchases, the government affects the net price ratio and the optimal decision of the firm. If $\frac{(1-\tau_P)}{(1-\tau_S)} > 1$, then the relative prices increase, firm optimal input level consumption decreases, firms use less M , and viceversa, a ratio less than one decreases relative prices leading firms to consume more inputs.

The fix: redefine the ratio, don't add a term

Solving Equation 3 for ε and regrouping the two logs into one ratio:

$$\varepsilon_{it}(M_{it}) = \ln \beta - \ln \frac{\rho_t M_{it} (1 - \tau_{P,it})}{P_t Y_{it} (1 - \tau_{S,it})} \quad (4)$$

— i.e. use the ratio of **net-of-tax nominal purchases to net-of-tax nominal sales** in place of the gross ratio, rather than tacking on a separate wedge term. Purely a repackaging of Equation 3 algebraically, but the better way to think about it: both legs of the ratio are still directly-observed nominal quantities times directly-observed tax rates — no price index, no deflator, nothing new to estimate. This means the “nominal log-share needs only the ratio, immune to price-index quality” result (CLAUDE.md, 2026-08-25 entry) survives completely unchanged; the wedge only ever needs $\tau_{S,it}, \tau_{P,it}$ themselves, both already columns in `stage2_data`.

Nothing downstream changes in functional form. $\mathcal{V}_{it} = u_{it} - \varepsilon_{it}$ is a definitional identity that holds for whichever ε is plugged in — it was never tied to the no-wedge case. Re-deriving $\mathcal{W}_{it} = y_{it} - \beta(m_{it} - \mathcal{V}_{it})$ with the wedge-corrected ε still collapses to $\alpha_K k_{it} + \alpha_L l_{it} + \omega_{it} + (1 - \beta)\varepsilon_{it}$ by the same algebra as before (the derivation only used the $\mathcal{V} = u - \varepsilon$ identity and the production identity, never the specific formula for $\varepsilon(M)$).

This is true because the identity $u = \ln\left(\frac{M^*}{M}\right)$ remains the same, we just have to add and subtract the log of the net after tax ratio of purchases to sales. I double-checked the algebra.

β first-stage results might be biased since the intercept of the regression captured $\ln \beta + E[\ln((1 - \tau_S)/(1 - \tau_P))]$. Deconvolution results might also change because any random variation of the ratio would have gone to ε_{it} . Since I use the density of ε to deconvolve the distribution of u , then f_u might be biased. **Do they change? How much? Have results ready in case supervisors ask. Review and analyze results.**

So `e_of_M`, `eps_of_M`, `omega_of_M`, `h_of_e` in `Code/Rcpp/1200-stage2-elvis.cpp` need **no changes**. Only two upstream data-construction pieces do:

1. **First-stage regression LHS** (`021-deconv-funs.R`, `first_stage_panel/first_stage_panel_me`): replace `var="log_mats_share"` with a new column

$$\text{log_mats_share_net} = \text{log_mats_share} + \ln(1 - \tau_P) - \ln(1 - \tau_S)$$

so the corp-only regression recovers $\ln \beta$ cleanly instead of $\ln \beta + \ln \frac{1 - \tau_S}{1 - \tau_P}$ (both τ_S, τ_P already exist as raw data columns — nothing new to construct beyond this one line). This is the piece that reopens stage 1, currently supervisor-approved/read-only per CLAUDE.md.

2. **$h(e)$'s tax input** (`1200-stage2-data.R` / the driver): use `sales_tax_rate_purchases` (τ_P) directly, replacing `tau_rho_proxy/eff_tax_rate_c`.

Status and scope

Not yet applied to any code — this note documents the derivation for the supervisor conversation, per the user's request, before touching `021-deconv-funs.R` or re-running stage 1. Because Equation 1 nests the currently-approved single- τ model exactly at $\tau_S = \tau_P$, this is a

strict generalization using a variable already sitting in the data, not a competing identification strategy — the “cost” is re-running stage 1 with the corrected regressor, not redesigning anything.

Maintained, not tested: τ_S, τ_P are treated as exogenous to the firm’s M/e choices (set by tax policy/product classification, not chosen by the firm). If firms could manipulate which goods they report selling to shift tax brackets, this would reintroduce endogeneity — no different in kind from the maintained exogeneity of ρ_t, P_t already relied on elsewhere, but worth flagging as an assumption rather than a proven fact.

This paper focuses on overreporting inputs, not underreporting sales. First, it is documented underreporting sales is more concentrated in retailers, B2C firms, than in B2B, most manufacturing firms, which is the focus of this paper. Second, firms selling to other firms face counter-incentives; the firms selling wants to underreport its sales, but the firm buying from selling firm wants to overreport the purchase, therefore they are not likely to cooperate, so we assume sales from manufacturing firms that sell to other firms do not underreport sales. In contrast, firms can buy fake invoices to claim additional deductions from sales taxes. The key difference is that there is one real transaction in the sales of products in which participants need to coordinate when firms underreport sales, in the case of firms selling to consumers, since consumers pay the final price and do not have incentives to deduct their purchases, firms might offer consumers a lower price if the consumer accepts not requesting an invoice. In the overreporting of inputs, there is no real transaction, no exchange of goods, only the price of paying someone (with a ghost firm, usually) a fee for fabricating fake invoices.

Government revenue and the corner solution, revisited

Building on the materials FOC above (Equation 3): the counterfactual’s revenue-maximizing purchases-credit multiplier Δ (600-opt-tax.qmd’s Counterfactual section) is found empirically to be pushed to the corner $\Delta = -1$, with no interior optimum appearing under simulation. This section shows *why*, for any admissible q, κ — not just the linear-linear headline specification — and then checks whether letting the true materials level respond to the tax rate (via Equation 3, held fixed in the counterfactual as implemented) can rescue an interior optimum.

Fixed M : no interior optimum, for any q, κ

Let $x \equiv 1 + \Delta$, $B(x) \equiv x\tau_P\rho_t$ the benefit shifter, and write the government’s per-firm revenue (the Counterfactual section’s own formula) as

$$R(x) = t[1 - B(x)][M + L(e^*(x))], \quad L(e) \equiv (1 - q(e))e$$

where $e^*(x)$ solves the firm's evasion FOC Equation 2, written generally as $B(x)h(e) = \kappa'(e)$ with $h(e) \equiv 1 - q(e) - q'(e)e = L'(e)$.

Differentiating and substituting the firm's own FOC ($B(x)h(e^*) = \kappa'(e^*)$) to eliminate $h(e^*)$:

$$\frac{dR}{dx} = -\tau_P \rho_t [M + L(e^*(x))] - \kappa'(e^*(x)) \frac{de^*}{dx} \quad (5)$$

Both terms are negative under mild regularity: $M > 0$, $L(e) \geq 0$ for $q \in [0, 1]$, $\kappa'(e) > 0$ (evasion costly at the margin), and $de^*/dx > 0$ — forced by the firm's own SOC together with $h(e^*) > 0$, which is itself forced by the FOC given $\kappa' > 0$ and $B(x) > 0$. **So $dR/dx < 0$ everywhere, for any q, κ satisfying these — a structural property of this revenue formula, not an artifact of the linear- q /linear- κ specification.** The mechanism: this revenue formula is a pure transfer between the government and evading firms; nothing in it attaches a real resource cost to x , so a revenue-maximizing planner is always pushed to the corner — a Diamond–Mirrlees-flavored result, and the formal counterpart of CLAUDE.md's empirical “ $\Delta = -1$ is the unconstrained mathematical revenue-maximizer” finding.

Letting M respond: does it help?

The materials FOC (Equation 3, with $\tau_P \rightarrow x\tau_P$ under the counterfactual) implies

$$M(x) \propto (1 - x\tau_P)^{-1/(1-\beta)}, \quad M'(x) = \frac{\tau_P}{1-\beta} \cdot \frac{M(x)}{1 - x\tau_P} > 0$$

— true materials usage *rises* with the credit rate: standard downward-sloping factor demand. A bigger credit lowers the net price $(1 - \tau_P)\rho_t$, so the firm, efficiently given its own concave production function, uses more of the now-cheaper input. (Matches this file's own line above, §Materials FOC: “a ratio less than one decreases relative prices leading firms to consume more inputs.”)

Since $Y = K^{\alpha_K} L^{\alpha_L} M^{\beta} e^{\omega+\varepsilon}$, this response propagates directly to $t1(x) = \tau_S P_t Y(x)$ — **not fixed once M moves, even though τ_S itself never changes** (an inconsistency in treating $t1$ as literally constant while letting M respond, caught and corrected in the derivation of this section):

$$\frac{dt1}{dx} = \tau_S P_t \beta \frac{Y}{M} M'(x) = \frac{\tau_S \rho_t (1 - x\tau_P)}{1 - \tau_S} M'(x) > 0$$

More real materials $\Rightarrow$ more output $\Rightarrow$ more sales-tax revenue. Adding this to Equation 5, the $(1 - x\tau_P)$ factors cancel cleanly between the two $M'(x)$ terms:

$$\frac{dR}{dx} = \rho_t M'(x) \cdot \frac{\tau_S - x\tau_P}{1 - \tau_S} - \tau_P \rho_t [M(x) + L(e^*(x))] - \kappa'(e^*(x)) \frac{de^*}{dx} \quad (6)$$

The new first term is **positive whenever $x\tau_P < \tau_S$** — below a threshold $x_c \equiv \tau_S/\tau_P$, extra credit-subsidized materials raises more sales-tax revenue than it costs in credit — and negative

above it. This is a genuine candidate for an interior root, unlike the fixed- M case: the sign of dR/dx is no longer settled a priori. (At $x = x_c$ exactly the first term is zero and the other two are still negative, so any interior root sits strictly below x_c , where the positive term must be *large*, not merely positive, to flip the sum.)

Numeric check: real, but four to five orders of magnitude too small to matter

Evaluated at the headline operating point ($\hat{\lambda} = 5.427 \times 10^{-7}$, moment set A, lag_m — CLAUDE.md) and sample medians from Code/Products/1200-stage2-data.RData (interior unincorporated firm-periods, $\tau_P > 0$): $\hat{\beta} = 0.387$, $\hat{\tau}_P = 0.0876$, $\hat{\tau}_S = 0.0888$, median $M^* \approx 8,508$, median $e \approx 841$. ($x_c = \tau_S/\tau_P \approx 1.014$ at the median firm — but only 34% of firm-periods have $\tau_S > \tau_P$ at all, so most firms have the wrong sign for this channel even before magnitude is considered.)

At $\Delta = 0$ ($x = 1$), median firm, the three terms of Equation 6 ($\rho_t = 1$, nominal COP throughout):

term	value
output-growth channel (new)	$\approx +1.8$
mechanical legit-credit channel	≈ -819
evasion-response channel	$\approx -80,655$

The evasion channel dominates the new channel by \$ 45,000×\$, the mechanical channel by \$ 450×\$ — both driven by $1/(2\hat{\lambda}) \approx 921,000$, vastly larger than any M or e scale in the data (the same mechanism already flagged in CLAUDE.md as “ λ is the dominant axis” on the estimation side). Checked also in the deep-cut, fully-clipped region ($x = 0.9$, $e^* = 0$ so the evasion term vanishes entirely): output-growth $\approx +14$ vs. mechanical ≈ -734 — still \$ 50×\$ apart.

Conclusion: the output-growth channel is real — correctly signed, correctly derived — but numerically negligible at the currently estimated $\hat{\lambda}$. It does not rescue an interior Δ^* , and the corner result stands. It would only start to matter if enforcement were estimated to be dramatically stronger (a much larger $\hat{\lambda}$, shrinking $1/(2\lambda)$ toward the same order as typical M) — worth a sensitivity re-check if $\hat{\lambda}$ ’s estimate ever moves substantially, but not a priority now.

Scope: this is still a local result

Both Equation 5 and Equation 6 describe how revenue responds to Δ **around the observed equilibrium**, holding $K, L, \omega, \varepsilon$ fixed throughout — a comparative-statics Laffer curve, not a general-equilibrium optimal-tax result. Letting true M (and hence Y) respond via its own

FOC is one step toward real economic effects of the tax change, but capital, labor, entry/exit, and cross-firm reallocation all remain frozen. The corner conclusion is robust to *this particular* extension at current estimates; it says nothing about whether a fuller general-equilibrium treatment (K/L margins, firm entry/exit) would change the answer. Not pursued further now — flagged as a future extension if the local result is ever challenged on these grounds, or if $\hat{\lambda}$'s estimate moves enough to make the output-growth channel numerically relevant.